

REMARKS ON THE SPORADIC RATIONAL TETRAHEDRA THAT MIGHT TILE SPACE

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ABSTRACT. Chentouf and Sun showed that at most 40 of the 59 sporadic rational tetrahedra of Kedlaya, Kolpakov, Poonen and Rubinstein can tile $\mathbb{R}^3$, and they left these 40 undecided. We make four remarks.

- (1) The printed list contains 42 tuples, two of which are excluded in the body of their paper.
- (2) Two members of the list are classical space-fillers, namely Sommerville's tetrahedra No. 2 and No. 3.
- (3) None of the remaining 38 tetrahedra admits a face-to-face tiling. For 12 of them this is new.
- (4) In any tiling by one of the 38, faults are forced along every copy of certain edges, and convex clusters of at most 7 copies never tile isohedrally.

We also explain why purely local arguments cannot settle the icosahedral members of the list.

1. SETTING

A tetrahedron is *rational* if all of its dihedral angles are rational multiples of π . Kedlaya, Kolpakov, Poonen and Rubinstein [5] classified such tetrahedra into two one-parameter families and 59 sporadic examples. Since rational tetrahedra have Dehn invariant zero, Debrunner's theorem [3] gives no obstruction to tiling. Chentouf and Sun [2] gave three necessary criteria:

- their Proposition 2.3, for tilings that are not face-to-face;
- their Proposition 2.7, for face-to-face tilings;
- a length-weighted linear program, their Corollary 2.13.

They proved that the sporadic tetrahedra that tile lie in a set $\mathcal{A}$ of 40 tetrahedra (their Theorem 1.2). Whether the members of $\mathcal{A}$ tile was left open.

We use their conventions. Vertices are labelled 1, 2, 3, 4, α_{ij} is the dihedral angle along edge ij , and angles are listed in the order $(\alpha_{12}, \alpha_{34}, \alpha_{13}, \alpha_{24}, \alpha_{14}, \alpha_{23})$ as multiples of π . A tiling is *face-to-face* (f2f) if no edge of a tile meets the relative interior of a face of another tile [2, Def. 2.2]. Tilings may use mirror images. All computations are available at [7] (directory `research/p2`).

2. THE LIST

We number the 42 tuples printed in [2, Thm. 1.2] as #1, ..., #42 in reading order. Tuples #34 and #35,

$$\left(\frac{3}{20}, \frac{11}{20}, \frac{11}{20}, \frac{1}{4}, \frac{1}{3}, \frac{2}{3}\right) \quad \text{and} \\ \left(\frac{11}{60}, \frac{31}{60}, \frac{11}{20}, \frac{1}{4}, \frac{3}{10}, \frac{7}{10}\right),$$

are exactly the two tetrahedra that [2, §3.1, eq. (5)] excludes by the linear program. So the printed set has 42 elements, and the intended $\mathcal{A}$ consists of the other 40.

We checked every tuple as follows:

- the Gram matrix $(-\cos \alpha)$ is singular to within 10^{-40} and positive semidefinite, with a positive kernel vector;

- the dihedral angles recomputed from the reconstructed vertices agree with the data to within 10^{-60} .

This guards against transcription errors. Our implementation of the linear program of [2, Cor. 2.13] excludes precisely #34 and #35 and is feasible for the other 40.

3. TWO CLASSICAL TILERS IN $\mathcal{A}$

Proposition 1. *Tetrahedron #1 = $(\frac{1}{4}, \frac{1}{3}, \frac{1}{3}, \frac{1}{4}, \frac{1}{2}, \frac{2}{3})$ is Sommerville's No. 3. Tetrahedron #3 = $(\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{3}, \frac{1}{2})$ is Sommerville's No. 2, that is, the Coxeter simplex $[4, 3^{1,1}]$. Both tile $\mathbb{R}^3$.*

Proof. In the notation of Table 1 of [1], Sommerville No. 3 has angles $2\pi/n_{ij}$ with

$$(n_{12}, n_{13}, n_{14}, n_{23}, n_{24}, n_{34}) = (3, 6, 6, 8, 8, 4).$$

Relabelling its vertices by $1 \mapsto 3, 3 \mapsto 1$ gives #1. Sommerville No. 2 has $(4, 4, 4, 6, 6, 8)$, and the relabelling $1 \mapsto 3, 2 \mapsto 4, 3 \mapsto 1, 4 \mapsto 2$ gives #3. A tetrahedron is determined up to similarity by its ordered dihedral angles.

Independently, we verified tilings with the Poincaré polyhedron theorem. For #3, the reflections in all four faces satisfy the edge conditions: every dihedral angle is π/m , so this is the Coxeter group $[4, 3^{1,1}]$. For #1, the following pairing satisfies all edge-cycle conditions exactly:

- the faces opposite vertices 1, 2, 3 are mirrors;
- the face opposite vertex 4 is paired with itself by the half-turn about its axis of symmetry.

In all, our search finds 6 isohedral face pairings for #1 and 4 for #3. $\square$

Hence at most 38 members of $\mathcal{A}$ are genuinely undecided.

4. NO FACE-TO-FACE TILINGS

Lemma 2. *In a face-to-face tiling by tetrahedra, any two tiles meet in a common face, edge or vertex, or not at all. In particular, the tiles containing a point in the relative interior of an edge E of a tile all have E as an edge, and consecutive tiles around E share a whole face.*

Proof. If two coplanar faces overlap but are not equal, the boundary of one meets the interior of the other [2, Lemma 2.6]. This is excluded, so overlapping coplanar faces coincide.

Let p lie in the relative interior of an edge E of a tile A . Let F be a face of A containing E . The tile across F near p has a face overlapping F , hence equal to F , so it contains E as an edge. Repeating this around E shows that every tile containing p has E as an edge. The remaining cases follow in the same way. $\square$

Definition 3. Fix an edge length ℓ . A *wedge* is a tile together with one of its edges ab of length ℓ , placed on a fixed segment PQ (with $a \mapsto P, b \mapsto Q$ or the reverse), and an ordering $(F_{\text{in}}, F_{\text{out}})$ of its two faces through ab . The angle of the wedge is α_{ab} .

A face through PQ is described by the pair $(|Px|, |Qx|)$, where x is its third vertex. The *star graph* of ℓ has these pairs as vertices and the wedges as arcs from F_{in} to F_{out} . An *edge star* of an edge e of T is a closed walk in the star graph of total angle 2π that uses a wedge of T on e .

Mirror images are taken into account automatically, because only metric data enter.

Proposition 4. *If T admits a face-to-face tiling, then every edge of T has an edge star.*

Proof. By Lemma 2, the tiles around a copy of an edge e form a cyclic sequence of wedges on that edge. Consecutive wedges share a whole face, so the face data match, and the angles sum to 2π . $\square$

Proposition 2.7 of [2] is the special case of Proposition 4 for tetrahedra whose edge lengths are pairwise distinct or parallelogram-like. In that case the star graph forces repeated reflections.

Theorem 5. *None of the 38 tetrahedra in $\mathcal{A} \setminus \{\#1, \#3\}$ admits a face-to-face tiling. For $\#2, 5, 7, 12, 19, 20, 22, 23, 25, 27, 36, 42$, this does not follow from [2, Prop. 2.7].*

Proof. For each of the 38 tetrahedra, at least one edge has no edge star; Table 1 lists these edges. Checking this is a finite search in a graph with at most 12 vertices, carried out over $\mathbb{Q}$ for the angles. As a control, every edge of $\#1$ and $\#3$ has an edge star.

Edge lengths are computed to 60 digits. Distinct length classes differ by at least 0.0093, and equal lengths agree to 10^{-59} . Wrongly identifying two lengths could only add arcs to the star graph, so the non-existence statements are robust under this identification.

For the last claim, the 12 listed tetrahedra are exactly those in which the edge lengths are neither pairwise distinct nor parallelogram-like, or in which every dihedral angle divides 2π . For these, Proposition 2.7 of [2] does not apply. $\square$

Remark 6. None of these tetrahedra can be excluded by Proposition 2.3 of [2], because each of the 40 has several dihedral π -combinations. So every potential tiling is necessarily non-face-to-face.

5. FORCED FAULTS

Call an edge of T *bad* if it has no edge star. A *metric chain* is a path in the star graph through a wedge of T on that edge; its angle is the sum of the wedge angles. For all 38 tetrahedra, no bad edge has a metric chain of total angle exactly π .

Take a tiling and a generic point p on a copy of a bad edge. The tiles around p split into maximal metric chains. Between two consecutive chains there is a *fault half-plane*, in which the faces of the two adjacent tiles are different triangles. Alternatively, p lies in the relative interior of a face of another tile. Hence:

Proposition 7. *In any tiling by one of the 38 tetrahedra, every point of every copy of a bad edge lies on a fault half-plane, or in the interior of a face of another tile. In particular, faults occur along a set of positive length in every tile.*

Clusters. A natural way to produce faults is to glue copies of T face-to-face into a convex polytope Q that tiles. Neighbouring copies of Q may then triangulate their common faces differently. Interior edges of Q need edge stars, and edges inside faces of Q need metric chains of angle π . Therefore every bad edge lies on an edge of Q , with Q 's dihedral angle there equal to a metric-chain angle.

We enumerated all clusters of at most 7 copies glued face-to-face (mirror images allowed), up to congruence. For each convex cluster we tested the Poincaré conditions over all pairings of its polygonal faces by isometries, including translations, half-turns, reflections and point reflections:

- for $\#1$ and $\#3$ the method finds many tiling clusters;
- for the other 38 there are only 0–24 convex clusters (last column of Table 1), and none tiles.

6. ICOSAHEDRAL MEMBERS

Many members of $\mathcal{A}$ are icosahedral. Take, for example, $\#19 = (\frac{1}{5}, \frac{1}{5}, \frac{2}{5}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3})$ and $\#25 = (\frac{1}{5}, \frac{1}{3}, \frac{1}{3}, \frac{1}{5}, \frac{1}{2}, \frac{4}{5})$.

- Every dihedral angle lies in $\{\frac{\pi}{5}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{5}\}$.
- Every vertex figure is a union of 1 to 14 copies of the Schwarz triangle $(2, 3, 5)$, which tiles the sphere. Vertex 4 of $\#19$ and vertices 1, 4 of $\#25$ are this triangle.
- The golden ratio appears among the edge lengths: $|24|/|23| = \varphi$ for both, and also $|12|/|23| = \varphi$ for $\#25$.

In both, the only bad edge is 23, of unique length.

Every local angular condition is satisfiable for these tetrahedra:

- the edge combinations around every edge;
- spherical vertex stars;
- planar vertex stars in fault planes, with the compatibility condition that $\pi - \alpha_g - \alpha_{g'}$ is a non-negative integer combination of dihedral angles across each interior edge;
- the length-weighted linear program, which has no forced combination.

An obstruction, if one exists, must therefore be global and metric. The distinct edge lengths are linearly independent over $\mathbb{Q}$ numerically (PSLQ finds no relation with coefficients up to 10^6). This suggests counting edge lengths along lines that carry faults.

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#	$(\alpha_{12}, \alpha_{34}, \alpha_{13}, \alpha_{24}, \alpha_{14}, \alpha_{23})/\pi$	edges w/o star	status	clusters
1	(1/4, 1/3, 1/3, 1/4, 1/2, 2/3)	–	tiles (Sommerville No. 3)	31
2	(5/24, 3/8, 1/3, 1/4, 13/24, 5/8)	12, 34, 14, 23	no f2f (new)	1
3	(1/4, 1/2, 1/2, 1/3, 1/3, 1/2)	–	tiles (Sommerville No. 2)	28
4	(7/24, 11/24, 13/24, 7/24, 1/3, 1/2)	12, 34, 13, 24	no f2f	3
5	(1/5, 1/5, 1/3, 1/5, 2/3, 2/3)	14, 23	no f2f (new)	0
6	(1/5, 1/5, 4/15, 4/15, 3/5, 11/15)	13, 24, 14, 23	no f2f	0
7	(1/5, 1/5, 1/3, 1/3, 3/5, 3/5)	14, 23	no f2f (new)	0
8	(1/3, 1/3, 3/5, 1/3, 2/5, 2/5)	13, 14, 23	no f2f	3
9	(4/15, 8/15, 1/3, 1/3, 7/15, 7/15)	12, 34, 14, 23	no f2f	2
10	(1/7, 3/7, 1/3, 1/3, 4/7, 4/7)	34, 14, 23	no f2f	0
11	(2/7, 2/7, 1/3, 1/3, 3/7, 5/7)	12, 34, 14, 23	no f2f	1
12	(1/5, 2/5, 1/2, 1/3, 1/3, 2/3)	34, 23	no f2f (new)	4
13	(2/15, 7/15, 1/2, 1/3, 2/5, 3/5)	12, 34, 14, 23	no f2f	2
14	(2/15, 7/15, 31/60, 19/60, 5/12, 7/12)	12, 34, 13, 24, 14, 23	no f2f	1
15	(1/5, 2/5, 7/12, 1/4, 5/12, 7/12)	34, 13, 14, 23	no f2f	1
16	(13/60, 23/60, 7/12, 1/4, 2/5, 3/5)	12, 34, 13, 14, 23	no f2f	1
17	(1/5, 3/5, 1/3, 1/3, 1/2, 1/2)	34	no f2f	4
18	(3/10, 7/10, 1/3, 1/3, 2/5, 2/5)	12, 34, 14, 23	no f2f	2
19	(1/5, 1/5, 2/5, 1/3, 1/2, 2/3)	23	no f2f (new)	5
20	(1/6, 7/30, 11/30, 11/30, 1/2, 2/3)	34, 13, 24, 23	no f2f (new)	5
21	(1/5, 1/5, 9/20, 17/60, 11/20, 37/60)	13, 24, 14, 23	no f2f	0
22	(1/5, 1/3, 1/2, 1/3, 2/5, 3/5)	14	no f2f (new)	6
23	(1/6, 11/30, 1/2, 1/3, 13/30, 17/30)	34, 14, 23	no f2f (new)	6
24	(1/6, 11/30, 29/60, 7/20, 5/12, 7/12)	34, 13, 24, 14, 23	no f2f	2
25	(1/5, 1/3, 1/3, 1/5, 1/2, 4/5)	23	no f2f (new)	3
26	(7/60, 5/12, 1/3, 1/5, 7/12, 43/60)	12, 34, 14, 23	no f2f	0
27	(1/5, 2/5, 2/5, 1/5, 1/2, 2/3)	34, 13	no f2f (new)	9
28	(1/5, 2/5, 1/3, 1/3, 1/2, 3/5)	34, 23	no f2f	3
29	(7/30, 13/30, 3/10, 3/10, 1/2, 3/5)	12, 34, 13, 24, 23	no f2f	3
30	(1/4, 7/20, 1/3, 1/3, 9/20, 13/20)	34, 14, 23	no f2f	2
31	(1/5, 1/2, 3/5, 1/5, 1/3, 2/3)	13, 23	no f2f	1
32	(1/5, 1/2, 17/30, 7/30, 3/10, 7/10)	13, 24, 14, 23	no f2f	1
33	(3/20, 11/20, 17/30, 7/30, 7/20, 13/20)	12, 34, 13, 24, 14, 23	no f2f	1
36	(1/5, 1/2, 1/2, 1/3, 2/5, 1/2)	14	no f2f (new)	24
37	(1/5, 1/2, 7/15, 11/30, 11/30, 8/15)	13, 24, 14, 23	no f2f	2
38	(4/15, 13/30, 17/30, 4/15, 2/5, 1/2)	12, 34, 13, 24, 14	no f2f	3
39	(4/15, 13/30, 3/5, 3/10, 11/30, 7/15)	12, 34, 13, 24, 14, 23	no f2f	3
40	(4/15, 17/30, 2/5, 3/10, 11/30, 8/15)	12, 34, 13, 24, 14, 23	no f2f	1
41	(3/10, 2/5, 3/5, 3/10, 1/3, 1/2)	12, 34, 13, 24	no f2f	3
42	(1/3, 2/5, 2/5, 1/3, 1/2, 2/5)	34, 13	no f2f (new)	10

TABLE 1. The 40 members of $\mathcal{A}$, numbered by position in the printed list of [2] (#34 and #35 omitted). “No f2f” means that no face-to-face tiling exists (Theorem 5); “new” means it is not covered by [2, Prop. 2.7]. “Clusters” is the number of convex face-to-face clusters of at most 7 copies, up to congruence.